## TOPIC 2 Transforming and Comparing Linear Functions

Name _____ Date _____

**Reminders**

Slope = $(y_2 - y_1) \div (x_2 - x_1)$ · $y = mx + b$ · point-slope: $y - y_1 = m(x - x_1)$

Parallel: same slope · Perpendicular: flip the fraction and change the sign (product $-1$)

Outside $f(x) + k$: up/down · Inside $f(x - h)$: right $h$, $f(x + h)$: left $h$ (opposite the sign)

Horizontal: $y = \#$, slope 0 · Vertical: $x = \#$, slope undefined (HOY VUX)

Domain = inputs (x) · Range = outputs (y) · Zero = x-intercept

**1.** What are the equation and slope of the vertical line through $(-7, 2)$?

**A.** $y = 2$; slope is zero

**B.** $x = -7$; slope is zero

**C.** $x = -7$; slope is undefined

**D.** $y = -7$; slope is undefined

**2.** Write the equation in slope-intercept form of the line through $(-1, 0)$ that is parallel to $y = 5x - 2$.

**3.** $f(x) = 2x + 1$ is translated 3 units left. What is the y-intercept of the new function?